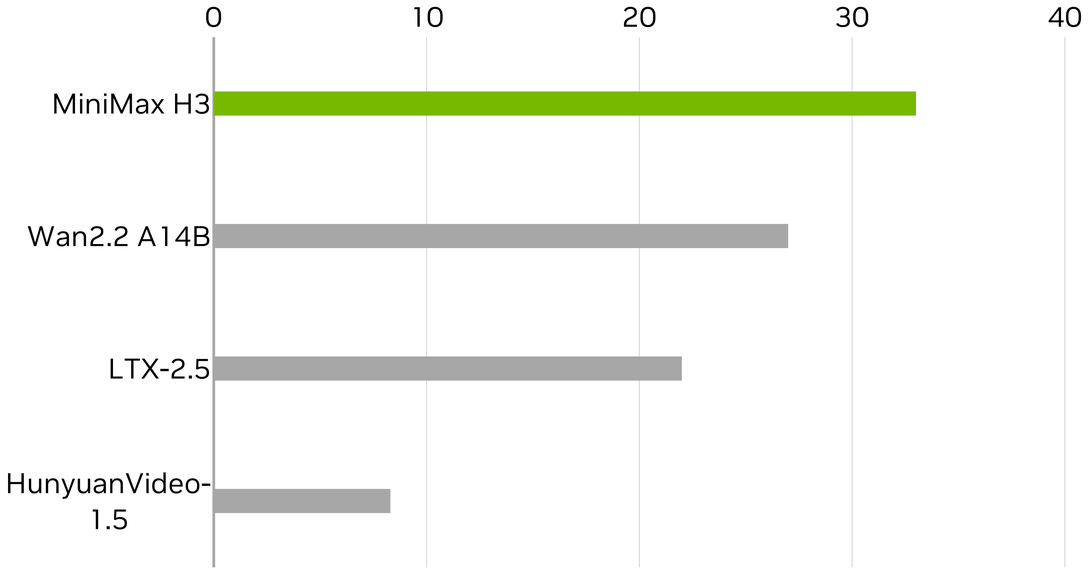
Toward Real-Time Generation With MiniMax H3

8× RTX PRO 5000 · vLLM-Omni

Shunyang Li | NVIDIA Devtech | 1 Nov 2026

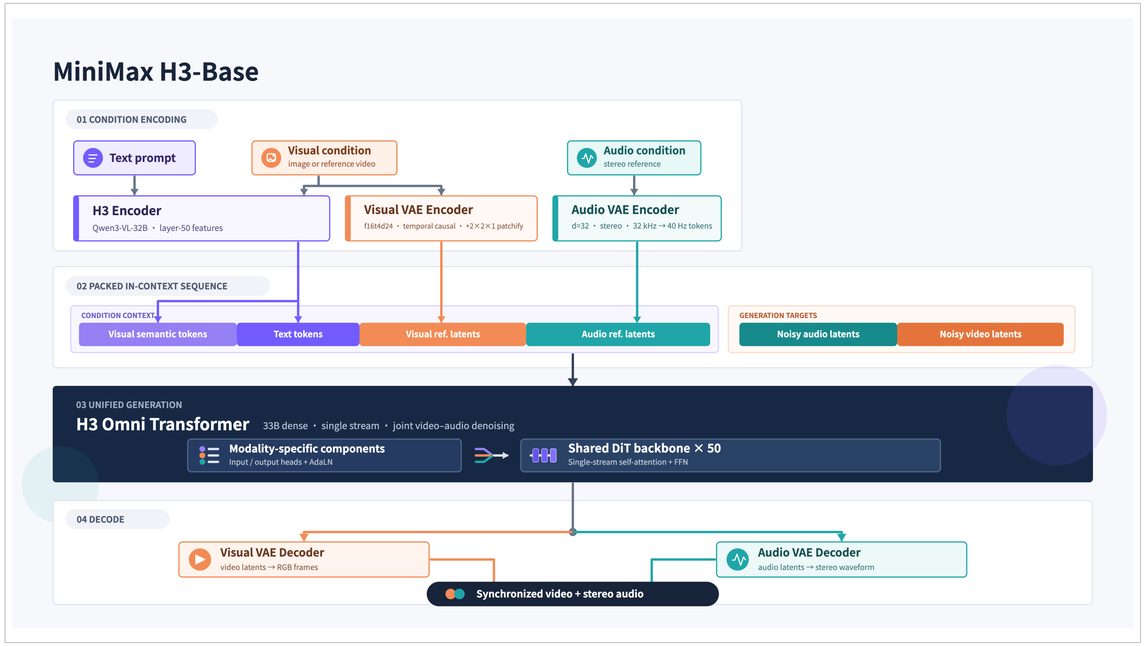
Real-Time Ref2VA Raises the Systems Bar

A 124-frame clip at 24 FPS is real time only when end-to-end latency stays below 5.167 s (RTF < 1)



Generative Backbone · Full Parameters (B)

RTF = latency ÷ media duration; the 124-frame deadline is 5.167 s. H3 has the largest dense backbone in this peer set; Wan2.2 has 27B total and 14B active parameters



Model Capability · Native 768P · 33B Dense

A 32B encoder feeds the 33B Omni Transformer and visual/audio VAEs. Fifty transformer blocks are evaluated at each of 50 baseline steps.

Sources: MiniMax H3, Wan2.2, LTX-2.5 and HunyuanVideo-1.5 official releases · author 960×544 Ref2VA workload

SM120 Cards

GPU	SMs	Memory	Power	Interconnect
RTX PRO 5000	110	72 GB GDDR7 ECC	300 W	PCIe Gen5 ×16 No GPU NVLink
RTX PRO 6000 **Workstation / Server**	188	96 GB GDDR7 ECC	600 W	PCIe Gen5 ×16 No GPU NVLink
GeForce RTX 5090	170	32 GB GDDR7	575 W	PCIe Gen5 ×16 No NVLink
DGX Spark **GB10 · SM121**	48	128 GB Unified LPDDR5X	140 W SoC	CPU–GPU NVLink-C2C 200 Gb/s ConnectX-7

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×8

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Baseline: Latency Breakdown

E2E: 213.402 s · Generated video: 960 × 544, 5.167 s

Reference configuration

- 8 × PRO 5000 GPUs (SM120)
- Original Ref2VA
- 50 steps · BF16 · dense attention
- DiT TP1 / Ulysses8
- Text encoder TP8 · VAE tiles8
- Native VAE · FP16 decode

GPU work by operator

	Mean s / GPU	Share
NCCL · includes waiting	122.182	**57.21%**
Attention	47.105	**22.06%**
GEMM	39.650	**18.57%**
Norm / activation / elementwise	2.288	1.07%
Layout / device copies	1.929	0.90%
CUDA memcpy / memset	0.064	0.03%
Convolution	0.192	0.09%
Other kernels	0.143	0.07%

Nsight GPU work includes overlap and waiting; shares exclude CPU work and idle gaps.

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Three Ideas Behind Faster Generation

Technical overview · detailed algorithms and source excerpts remain available on hidden slides

Less model work

Few-step distillation reduces evaluations.
Sparse attention concentrates work on selected regions.
Quantized inference lowers arithmetic and memory cost.

Overlap across eight GPUs

Ulysses distributes attention across heads.
RDMA carries the exchanges.
Head bands overlap transfer with attention compute.

Keep the output path moving

CUDA Graph replays stable head-band compute.
Fused packing reduces intermediate copies.
NVFP4 VAE and MP4 pre-encoding reduce output cost.

These are optimization families. Use each demo’s own recipe and measurements; the four-step avatar and slide 16 are distinct workloads.

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Optimization Path: E2E Speedup

960 × 544 · 5.167 s output · 8 GPUs · standalone H3

Stage	Cumulative change	Median (s)	E2E speedup vs. previous	RTF
00	Baseline · original BF16 / dense / 50 evaluations	213.402	—	41.304
01	+ PDD · 50 → 8 evaluations	44.327	4.814×†	8.579
02	+ Trained VSA · native 8-step model / runtime	25.915	1.710×	5.016
03	+ QAD NVFP4 DiT / native FP4 attention	19.460	1.332×	3.767
04	+ RDMA / direct layout / V-gate overlap	6.832	2.848×	1.322
05	+ 2 / 3 / 2 head pipeline	6.407	1.066×	1.240
06	+ Compute CUDA graphs	6.193	1.035×	1.199
07	+ 36-layer NVFP4 VAE / MP4 pre-encoding	5.420	1.143×	1.049
08–12	**+ Output and data-movement optimizations**	**4.922**	**1.101×**	**0.953**

08–12: fused tile / AAC preparation, direct I/O packing, compact routing and Q/K/V copy removal.

E2E speedup = previous displayed median ÷ current median. 08–12 compares the final result with stage 07.

Final median: 4.922 s · recheck: 4.927 s · 6/6 measured requests below 5 s

† Recorded median ratio; baseline refreshed separately (n = 2). Stages 01–12 retain n = 3 each; no isolated 50→8-step gain claimed.

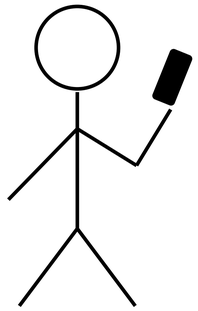
RTF = latency / 5.167 s. Standalone H3; ASR / LLM excluded. Model, schedule and precision changes are not all lossless.

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Live Avatar: From a Question to a Video

A spoken question becomes a visible, audible reply.

“Hi, could you introduce yourself?”



User

Voice

Qwen3-ASR

0.6B · Speech → text

RECOGNIZED TEXT

“Hi, could you introduce yourself?”

Qwen3-4B

Reply → H3 prompt

ENGLISH PROMPT · EXCERPT

… “Hi, I’m Miles. I help with simple, clear answers.” … His lips follow the words.

Reply text + scene instructions

Reference image



H3

Audio + video

“Hi, I’m Miles…”



Avatar reply

Video + voice

Hear the avatar’s reply

960 × 544 · ~5 s per window · 8 × RTX PRO 5000

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Continuation: Two Clips, One Timeline

The previous ending guides the next clip. Only new content is appended.

CHUNK A / LAST FRAME



CHUNK B / FIRST RETAINED FRAME



CONTINUOUS PLAYBACK

Trim B’s overlapping start

Chunk A · 5.167 s | Chunk B · +4.958 s

Recent audio–video context conditions the next generation; matching audio is trimmed with the overlap.

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Live Avatar: Recorded Conversation

74.55 seconds · user-recorded microphone conversation · native H3 audio + video



Play the recorded interaction · fullscreen avatar, live captions and timing overlays

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Takeaways

01 Multi-GPU optimization starts with topology

Map GPU/NIC links and locate communication bottlenecks.
Use available bandwidth efficiently, or reduce data transfer.
Communication is a key source of lossless speedups.

02 Evaluate quantization case by case

Lossy quantization behaves differently across models and workloads.
Measure quality and latency on representative cases.

03 Next: real-time response and richer interaction

Autoregressive diffusion for lower response latency.
Prompt enhancers for richer interaction.

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